

MAHDI HACHICHA

Embedded Systems | IoT | Robotics Engineering Student

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PROFESSIONAL SUMMARY

Computer Systems Engineering student (5-year integrated Engineering Diploma, expected 2027) specialised in embedded systems, IoT and robotics. Hands-on experience in ESP32 firmware and multi-protocol IoT gateways (Modbus, MQTT, LoRa, ESP-NOW), CAN bus / OBD-II automotive diagnostics, and ROS 2 autonomous navigation with LiDAR perception. Ranked 1st in class in years 3 and 4, with three consecutive Best Year-End Project awards.

EDUCATION

MedTech University (South Mediterranean University) — ABET accredited — Tunis, Tunisia

Engineering Diploma in Computer Systems Engineering (5-year integrated program) expected graduation June 2027

- Specialization: embedded systems, networks, IoT, robotics, low-level programming and systems architecture
- Ranked 1st in class in 3rd and 4th year

Integrated Preparatory Classes (MPI), 2022 – 2024 — mathematics, physics, computer science and engineering sciences

TECHNICAL SKILLS

Programming: C, C++, Python, Java, Verilog

Embedded Systems: ESP32, STM32, Arduino, Raspberry Pi 4, FPGA (Verilog), embedded web interfaces, embedded / edge AI

Hardware & Electronics: PCB design (Altium Designer), 3D mechanical design (Fusion 360), sensors and actuators, LiDAR

Communication Protocols: CAN bus, OBD-II, Modbus RTU/TCP, UART, I2C, SPI, MQTT, LoRa, ESP-NOW, Wi-Fi, Bluetooth, GSM

Robotics: ROS 2 (nodes, SLAM, trajectory planning, autonomous navigation), Gazebo, RViz 2, LiDAR perception

Software & Tools: Git, MATLAB / Simulink, Oracle APEX

Domains: Embedded systems, IoT, autonomous robotics, automotive diagnostics, embedded AI

PROFESSIONAL EXPERIENCE

Embedded Systems Intern — Tekkim, Tunis, Tunisia

June – July 2026 | ESP32, C/C++, Modbus RTU/TCP, MQTT, LoRa, ESP-NOW, GSM, Python

- Designed and developed a multi-protocol IoT gateway on ESP32, bridging Modbus RTU/TCP, Wi-Fi, Bluetooth, MQTT, GSM, LoRa and ESP-NOW within a single embedded firmware stack.
- Built an embedded web interface hosted directly on the ESP32, enabling on-device configuration of gateway and protocol settings without external tooling.
- Developed a Python application to simulate and monitor Modbus registers, enabling end-to-end testing and validation of the gateway without physical field devices.

Research Intern, Autonomous Vehicles — MedTech SmartLab, Tunis, Tunisia

2024 | ROS 2, LiDAR, Raspberry Pi 4, SLAM

- Developed a complete autonomous vehicle stack under ROS 2 on Raspberry Pi 4, integrating LiDAR perception, SLAM and trajectory planning for navigation in a controlled environment.
- Implemented the ROS 2 architecture end to end, covering inter-node communication and multi-sensor management.
- Validated the system on a real test course, achieving an obstacle detection rate above 95%.

Software Engineering Intern — SmartSystem Tunisia, Tunis, Tunisia

2023 | Oracle APEX, process digitalization

- Developed an interactive Oracle APEX interface digitalizing the management of banking declaration forms.
- Delivered the application to production for internal use by the Central Bank of Tunisia.

PROJECTS & VENTURES

PillPal — Founder & Hardware / 3D Design Lead

2026 – Present | ESP32, embedded AI, mobile application, 3D design

- Founded a startup around a smart medication dispenser, scaling an award-winning MedTech end of the year project (1st place, 2024) into a standalone company.
- Architected the device around an ESP32, automating morning, midday and night doses across a full week, fully configurable through the companion application.
- Integrated an AI-based feature in the companion app to detect whether a pill has been taken and notify the user.
- Led hardware architecture and 3D design of the device enclosure and dispensing mechanism.

Automotive Diagnostic System — Bako Motors

2025 | *CAN bus, OBD-II, Python, cloud dashboard*

- Designed and developed an embedded vehicle diagnostic system based on the CAN bus protocol with a custom PCB.
- Implemented a cloud-connected Python interface for real-time OBD-II monitoring and diagnostics.
- Integrated alerts and visualizations to support fault detection and preventive maintenance.

MiniMakers — Founder & Instructor, Robotics School (ages 9–15)

2025 – Present | *Arduino, robotics education*

- Founded and run a robotics school for children and teenagers, teaching 30+ students during a summer programme covering Arduino programming and hands-on robotics.
- Designed and led the flagship build: a four-wheel autonomous car with obstacle detection.

AWARDS & ACHIEVEMENTS

- Ranked 1st in class — 3rd and 4th year, Computer Systems Engineering, MedTech University
- Best Year-End Project, MedTech University — 2026, 2025 and 2024 (three consecutive years)
- 2nd Place — TechRevive 2.0, 2025
- 3rd Place — SMU AI Hackathon, 2025

CERTIFICATIONS

- PCB Design with Altium Designer — certified
- Robot Operating System 2 (ROS 2)

LEADERSHIP & AFFILIATIONS

- Founding Member, IEEE-HKN (Eta Kappa Nu) Honor Society, Nu Tau Chapter, SMU — since January 2026
- Treasurer, IEEE SMU Student Branch — 2024
- Treasurer, IEEE Robotics & Automation Society (RAS), SMU Student Branch — 2023

LANGUAGES

English — Fluent | French — Fluent | Arabic — Native